Yu-Chen Huang

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Profile

My research focuses on fast radio bursts (FRBs), which are millisecond-duration, extremely bright bursts of radio emission originating from cosmological distances. I am particularly interested in how FRBs propagate through and interact with astrophysical plasmas.

Education

- **University of Science and Technology of China** — *Sep 2022 - Present*
  PhD in Astrophysics (Integrated MSc–PhD Program) — Hefei, China
- **Sichuan University** — *Sep 2018 - Jun 2022*
  Bachelor in Physics — Chengdu, China

Publications

Primary Publications

[1] **Huang, Y.-C.**, Wang, J.-S., Zhang, S.-B., & Dai, Z.-G. 2026, Depolarization Induced by Rapid Polarization Angle Swings: A Common Feature of Pulsars and Fast Radio Bursts?, **Monthly Notices of the Royal Astronomical Society, 551, stag1457**.

[2] **Huang, Y.-C.**, & Dai, Z.-G. 2025, The Extreme Faraday Effect in Fast Radio Bursts, **The Astrophysical Journal Letters, 983, L24**.

[3] **Huang, Y.-C.**, & Dai, Z.-G. 2024, Fast Radio Bursts with Narrow Beaming Angles Can Escape from Magnetar Magnetospheres, **The Astrophysical Journal, 975, 226**.

[4] **Huang, Y.-C.**, Zhong, S.-Q., & Dai, Z.-G. 2024, Scattering Cross Sections of Magnetized Particles within Intense Electromagnetic Waves: Application to Fast Radio Bursts, **The Astrophysical Journal, 966, 97**.

Collaborative Publications

[1] An, T., Wang, A., **Huang, Y.-C.**, Feng, J.-P., et al. 2025, A Deep VLA Search for a Persistent Radio Counterpart to the One-off FRB 20250316A, **The Astrophysical Journal Letters, 991, L20**.

[2] Liu, Z.-N., Wang, W.-Y., **Huang, Y.-C.**, & Dai, Z.-G. 2025, Tera-Hertz Counterparts to Fast Radio Bursts from Coherent Cherenkov Radiation by Tilted Bunches, **The Astrophysical Journal, 989, 126**.

Academic Presentations

- **Fast Radio Burst 2026 Meeting** — *8 July 2026*
  Flash Talk (3 min) — Guiyang, China
- **FAST/Future Pulsar Symposium 15** — *3 July 2026*
  Oral Presentation (15 min) — Guiyang, China
- **The 3rd Chinese Fast Radio Burst Symposium** — *6 Jun 2025*
  Flash Talk (5 min) — Beijing, China
- **Fast Radio Burst 2024 Meeting** — *6 Nov 2024*
  Flash Talk (3 min) — Khao Lak, Thailand
- **The 2024 Annual Meeting of the Chinese Astronomical Society** — *31 Oct 2024*
  Oral Presentation (10 min) — Hangzhou, China
- **The 2nd Chinese Fast Radio Burst Symposium** — *5 Jun 2024*
  Oral Presentation (15 min) — Wuhan, China

Teaching Experience

- **The Physics of Compact Objects** — *Sep 2026 – Present*
  University of Science and Technology of China — Teaching Assistant
- **Radiative Processes in Astrophysics** — *Sep 2025 – Jan 2026*
  University of Science and Technology of China — Teaching Assistant
- **Introduction to Astrophysics** — *Mar 2021 – Jul 2021*
  Sichuan University — Teaching Assistant

Honors and Awards

- **Selected for the National Advanced Talent Cultivation Center for Physics of USTC** — *2025*
  University of Science and Technology of China
- **National Scholarship** — *2024*
  Ministry of Education of the People's Republic of China
  ◦ One of the most prestigious national scholarships for graduate students in China
- **First-Class Graduate Academic Scholarship** — *2022, 2023, 2024, 2025, 2026*
  University of Science and Technology of China
- **Selected for the Wu Yuzhang Honors College** — *2018*
  Sichuan University
  ◦ Top approximately 1.5% selected annually from undergraduate cohort

Skills

- **Programming Languages:** Wolfram Mathematica, LaTeX, Python

References

1. **Prof. Zi-Gao Dai**
   Professor, Academician of the Chinese Academy of Sciences
   Department of Astronomy
   University of Science and Technology of China
   Email: daizg@ustc.edu.cn
   Relationship: PhD Advisor